Reactivity & redox

Order the metals

1 Reactivity order

K > Na > Li > Ca > Mg > Al > C > Zn > Fe > H > Cu > Ag > Au. C and H are comparison points.

More reactive → greater tendency to lose electrons and form positive ions. Aluminium has a protective oxide layer.

2 Water & acid evidence

Cold water: K/Na/Li react readily; Ca reacts; Mg very slow; Zn/Fe/Cu no significant reaction.

Water products: hydroxide + H_2. Dilute HCl/H_2SO_4: Mg rapid, Zn slower, Fe slower still; Cu no reaction.

Compare under controlled concentration, temperature and surface area.

Follow electrons

3 Displacement

A more reactive metal displaces a less reactive one. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$.

Ionic: $\mathbf{Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu.}$ Sulfate spectator ions cancel.

Copper deposit; blue solution fades. Use several results to deduce an unknown order.

4 OIL RIG

Oxidation Is Loss; Reduction Is Gain of electrons. $\mathbf{Zn \rightarrow Zn^{2+} + 2e^-:}$ oxidation.

$\mathbf{Cu^{2+} + 2e^- \rightarrow Cu:}$ reduction. Atoms and total charges balance.

Metal + acid: metal oxidised; $2H^+ + 2e^- \rightarrow H_2$ is reduction.

Extract the element

5 Oxygen definitions

Oxidation gains oxygen; reduction loses oxygen.

$\mathbf{2CuO + C \rightarrow 2Cu + CO_2:}$ CuO reduced; carbon oxidised. Redox changes happen together.

Copper oxide loses oxygen; do not say the copper ions lose electrons.

6 Choose extraction

Unreactive gold may be found as metal. Most metals occur in compounds.

Metal below carbon: carbon can reduce its oxide. More reactive metal, or one that reacts with carbon: electrolysis may be used.

Melting and electrical current require energy; evaluate cost using supplied data.